

# Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: CRISISBENCH | Context: Argentina — Buenos Aires Policy Maker Cohort (Disaster Reporting Reliability)

**Input Content definition:** Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

Input content is the highest-priority dimension (HIGH per elicitation) and exhibits the most severe alignment failure. English tweets constitute 94.46% of the informativeness set [Q46], with non-English content largely confined to CrisisLex [Q47], and all classification experiments restricted to English [Q59]. The geographic distribution is concentrated in North American and globally reported events: Joplin tornado, Hurricane Sandy [Q27, Q28], 2010/2012 events [Q29], 2017 international disasters [Q31] — with no Argentine or Latin American disaster events documented. Dataset analysis empirically confirms that across 271 sampled examples, zero contain Argentine events, Argentine institutional actors, or Rioplatense Spanish; the only Spanish-language content is Chilean and Venezuelan [DATASET-D2, D3, D4, D6, D7]. The deployment's primary channel — Argentine Twitter/X with Rioplatense voseo, lunfardo, political satire, and references to local actors (SINAGIR, AFE, Buenos Aires province) — is entirely absent from the training distribution. The 2010–2017 temporal window [Q100] predates the dissolution of Télam (July 2024) [WEB-9], creation of AFE (2025) [WEB-10], and post-Musk Twitter/X conventions, meaning the institutional credibility landscape encoded in the data does not match the current Argentine deployment context.

| ID    | Evidence                                                                                                                                                                                                                                                                                                                                                                   |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q27   | "SWDM2013 dataset consists of data from two events: (i) the Joplin collection contains tweets from the tornado that struck Joplin, Missouri on May 22, 2011; (ii) The Sandy collection contains tweets collected from Hurricane Sandy that hit Northeastern US on Oct 29, 2012 (Imran et al. 2013a)." (p.3)                                                                |
| Q28   | "ISCRAM2013 dataset consists of tweets from two different events occurred in 2011 (Joplin 2011) and 2012 (Sandy 2012). Note that this set of tweets are different than SWDM2013 set even though they are collected from same events (Imran et al. 2013b)." (p.3)                                                                                                           |
| Q29   | "Disaster Response Data (DRD) consists of tweets collected during various crisis events that took place in 2010 and 2012. This dataset is annotated using 36 classes that include informativeness as well as humanitarian categories." (p.3)                                                                                                                               |
| Q31   | "CrisisMMD is a multimodal dataset consisting of tweets and associated images collected during seven disaster events that happened in 2017 (Alam et al. 2018). The annotations for this dataset is targeted for three classification tasks: (i) informative vs. not-informative, (ii) humanitarian categories (eight classes) and (iii) damage severity assessment." (p.3) |
| Q46   | "Among different languages of informativeness tweets, English tweets appear to be highest in the distribution compared to any other language, which is 94.46% of 156,899." (p.5)                                                                                                                                                                                           |
| Q47   | "Note that most of the non-English tweets appear in the CrisisLex dataset." (p.5)                                                                                                                                                                                                                                                                                          |
| Q59   | "Although our consolidated dataset contains multilingual tweets, we only use tweets in English language in our experiments." (p.6)                                                                                                                                                                                                                                         |
| Q100  | "The resulting dataset covers a time-span starting from 2010 to 2017, which can be used to study temporal aspects in crisis scenarios." (p.9)                                                                                                                                                                                                                              |
| D2    | RT @FlorGo: #FuerzaChile ,un 2 de abril ... Que vengan los ingleses a ayudarlos ahora ... A la larga o a la corta ,todo vuelve — Chilean Spanish earthquake solidarity tweet — not Argentine/Rioplatense                                                                                                                                                                   |
| D3    | Que lindo saber que en mi ciudad bella #Iquique hay gente con un corazón enorme dispuestos a ayudar... Emocionada — Chilean Spanish (Iquique city) solidarity tweet — Chilean, not Argentine                                                                                                                                                                               |
| D4    | #FuerzaChile jaja y piden ayuda jaja que los ayuden los ingleses manga de traidores — Chilean Spanish tweet expressing contempt for requesting outside help — not Argentine                                                                                                                                                                                                |
| D6    | IMPRESIONANTE terremoto Chile camara de seguridad farmacia - NEW VIDEO E...: http://t.co/bkUEIKfUj via @YouTube — Chilean Spanish earthquake tweet — Chilean Spanish, not Rioplatense                                                                                                                                                                                      |
| D7    | RT @marieli_d: Nuevamente se incendia #Amuay http://t.co/AiWlhgy3 — Venezuelan Spanish tweet about refinery explosion                                                                                                                                                                                                                                                      |
| WEB-9 | Télam dissolved by Decree 548/2024 in July 2024 — material change to Argentine formal press source landscape post-benchmark (https://www.boletinoficial.gob.ar/detalleAviso/primera/309815/20240701)                                                                                                                                                                       |

|        |                                                                                                                                                                                                    |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WEB-10 | AFE created by Decree 225/2025 — new federal emergency coordinator absent from benchmark institutional context ( <a href="https://www.mhewc.org/argentina/">https://www.mhewc.org/argentina/</a> ) |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Strengths

- Multi-event, multi-geography training distribution (20+ events spanning floods, earthquakes, hurricanes, typhoons, disease outbreaks, industrial accidents) provides exposure to diverse disaster registers, reducing categorical failure on hazard types absent from any single source [DATASET-D25, D27, D44]
- Some non-English Romance and Asian-language content (Italian, French, Portuguese, Tagalog, Spanish) is present and annotated, demonstrating the label schema has been exposed to multilingual data even though classification experiments are English-only [Q42, Q43, DATASET-D21, D39]

| ID  | Evidence                                                                                                                                                                                                                        |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q42 | "Some of the existing datasets contain tweets in various languages (i.e., Spanish and Italian) without explicit assignment of a language tag." (p.5)                                                                            |
| Q43 | "In addition, many tweets have codeswitched (i.e., multilingual) content." (p.5)                                                                                                                                                |
| D21 | RT @frankowolf1: il comune di Mirandola cerca ing. e arch. contattare Polizia Municipale: 0535/611039, 800/197197 #terremoto — Italian tweet seeking engineers after earthquake — non-English caution                           |
| D25 | After deadly Brazil nightclub fire, safety questions emerge. <a href="http://t.co/ah4JK2v2">http://t.co/ah4JK2v2</a> — Brazil nightclub fire — closest geographic proximity to Argentina                                        |
| D27 | Torrential rains have damaged Kenya's infrastructure, severing road links between Nairobi and Garissa, where the air base of the World Food Programme (WFP) is located. — Africa infrastructure damage — internationally framed |
| D39 | @lucilledizon: 517 ilang ilang st. bayanihan vilage cainta rizal, David Bautista +63 905 826 5094. Needs rescue @MMDA @govph #rescuePH — Tagalog rescue request — non-English, Philippine event, clear label                    |
| D44 | Colorado floods shut down hundreds of oil and gas wells; recovery will take time: Colorado's oil and... <a href="http://t.co/HivwsVtrc8">http://t.co/HivwsVtrc8</a> #Oil #BRK — North American oil/gas infrastructure damage    |

## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                                        |
|------|----------------------------------------------------------------------------|
| IC-1 | Do inputs require region-specific cultural/geographic/dialectal knowledge? |
| IC-2 | Does culturally sensitive content align with target culture?               |
| IC-3 | Flag inputs assuming Western-specific knowledge                            |
| IC-4 | Regional annotator recruitment for culturally sensitive instances          |
| IC-5 | Document content issues harming content validity                           |

### Checklist responses:

**IC-1:** Yes — Argentine deployment requires Rioplatense Spanish lexical and pragmatic competence (voseo, lunfardo, political satire, local institutional actors). The benchmark contains no Argentine content [DATASET CRITICAL Concern 1]; the only Spanish content is Chilean/Venezuelan [DATASET-D2, D3, D4, D6, D7].

**IC-2:** Cultural alignment is poor for social media (the highest-priority channel per elicitation Q2). Argentine political satire of government crisis response, lunfardo register, and references to AMBA institutional actors are absent. Formal press content may align better but is also unrepresented, and the dissolution of Télam materially changes the post-2024 source landscape [WEB-8, WEB-9].

**IC-3:** Yes — the benchmark embeds North American institutional and cultural framing (US sports commentary, US political discourse, Sandy/Joplin/Maria contexts) [DATASET-D17, D18, D46]. The vernacular register learned as 'noise' is specifically North American social noise, not Argentine.

**IC-4: INSUFFICIENT DOCUMENTATION** — no regional annotators were involved in constructing CrisisBench. Recruiting Argentine annotators to flag culturally sensitive instances would require a separate validation effort.

**IC-5:** Documented content issues: (a) zero Argentine/Rioplatense content; (b) only Chilean/Venezuelan Spanish at small scale [WEB-16, WEB-17]; (c) 2010–2017 temporal window predates current Argentine institutional landscape [Q100, WEB-9, WEB-10]; (d) North American event and institutional dominance. These violate content validity for the deployment's primary social media channel.

| ID     | Evidence                                                                                                                                                                                                                                                                                                                                                                   |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q2     | "We provide benchmarks for both binary and multiclass classification tasks using several deep learning architectures including, CNN, fastText, and transformers." (p.1)                                                                                                                                                                                                    |
| Q27    | "SWDM2013 dataset consists of data from two events: (i) the Joplin collection contains tweets from the tornado that struck Joplin, Missouri on May 22, 2011; (ii) The Sandy collection contains tweets collected from Hurricane Sandy that hit Northeastern US on Oct 29, 2012 (Imran et al. 2013a)." (p.3)                                                                |
| Q31    | "CrisisMMD is a multimodal dataset consisting of tweets and associated images collected during seven disaster events that happened in 2017 (Alam et al. 2018). The annotations for this dataset is targeted for three classification tasks: (i) informative vs. not-informative, (ii) humanitarian categories (eight classes) and (iii) damage severity assessment." (p.3) |
| Q46    | "Among different languages of informativeness tweets, English tweets appear to be highest in the distribution compared to any other language, which is 94.46% of 156,899." (p.5)                                                                                                                                                                                           |
| Q47    | "Note that most of the non-English tweets appear in the CrisisLex dataset." (p.5)                                                                                                                                                                                                                                                                                          |
| Q59    | "Although our consolidated dataset contains multilingual tweets, we only use tweets in English language in our experiments." (p.6)                                                                                                                                                                                                                                         |
| Q100   | "The resulting dataset covers a time-span starting from 2010 to 2017, which can be used to study temporal aspects in crisis scenarios." (p.9)                                                                                                                                                                                                                              |
| D2     | RT @FlorGo: #FuerzaChile ,un 2 de abril ... Que vengan los ingleses a ayudarlos ahora ... A la larga o a la corta ,todo vuelve — Chilean Spanish earthquake solidarity tweet — not Argentine/Rioplatense                                                                                                                                                                   |
| D3     | Que lindo saber que en mi ciudad bella #Iquique hay gente con un corazón enorme dispuestos a ayudar... Emocionada — Chilean Spanish (Iquique city) solidarity tweet — Chilean, not Argentine                                                                                                                                                                               |
| D4     | #FuerzaChile jaja y piden ayuda jaja que los ayuden los ingleses manga de traidores — Chilean Spanish tweet expressing contempt for requesting outside help — not Argentine                                                                                                                                                                                                |
| D6     | IMPRESIONANTE terremoto Chile camara de seguridad farmacia - NEW VIDEO E...: http://t.co/bkUEIKfUj via @YouTube — Chilean Spanish earthquake tweet — Chilean Spanish, not Rioplatense                                                                                                                                                                                      |
| D7     | RT @marieli_d: Nuevamente se incendia #Amuay http://t.co/AiWlhgy3 — Venezuelan Spanish tweet about refinery explosion                                                                                                                                                                                                                                                      |
| D17    | Tornado watch until 3am. Of course we just got our new garage door today. *sigh* — Personal complaint about tornado watch — Joplin 2011, North American event                                                                                                                                                                                                              |
| D18    | RT @tarnsnastylad: They need to name hurricanes better!.. superstorm Sandy? Seriously??.. Sounds like a FUCKING gay wrestler! — Crude personal opinion during Sandy — North American vernacular                                                                                                                                                                            |
| D46    | RT @eclecticbrotha: @AlGiordano Damn, I forgot how much of a shitshow Bernie's Puerto Rico operation was. — Political commentary during Hurricane Maria                                                                                                                                                                                                                    |
| WEB-8  | <a href="https://en.wikipedia.org/wiki/T%C3%A9lam">https://en.wikipedia.org/wiki/T%C3%A9lam</a>                                                                                                                                                                                                                                                                            |
| WEB-9  | Télam dissolved by Decree 548/2024 in July 2024 — material change to Argentine formal press source landscape post-benchmark ( <a href="https://www.boletinoficial.gob.ar/detalleAviso/primera/309815/20240701">https://www.boletinoficial.gob.ar/detalleAviso/primera/309815/20240701</a> )                                                                                |
| WEB-10 | AFE created by Decree 225/2025 — new federal emergency coordinator absent from benchmark institutional context ( <a href="https://www.mhewc.org/argentina/">https://www.mhewc.org/argentina/</a> )                                                                                                                                                                         |
| WEB-16 | CrisisLex T6 used crowdsourced workers for relatedness labels; no Argentine annotators documented ( <a href="https://crisislex.org/data-collections.html">https://crisislex.org/data-collections.html</a> )                                                                                                                                                                |
| WEB-17 | Sánchez et al. 2022 cross-lingual crisis classification paper does not address Rioplatense register ( <a href="https://arxiv.org/abs/2209.02139">https://arxiv.org/abs/2209.02139</a> )                                                                                                                                                                                    |

## Information Gaps

- Whether classifiers pre-trained on this benchmark transfer at any usable F1 level to Rioplatense Spanish content
- Whether the small Chilean subset would provide any positive transfer signal for Argentine deployment

## Requires Expert Verification

- Argentine linguist/policy expert validation of which Rioplatense register features (voseo, lunfardo, political satire patterns) most threaten classification accuracy
- Operational assessment by AMBA emergency management staff of how WhatsApp and Telegram channel content (not in benchmark) interacts with the Twitter/X channel monitoring

## Remediation

**Gap 1: Zero Argentine/Rioplatense Spanish content; 2010–2017 temporal window predates current Argentine institutional landscape (post-Télam, post-AFE)**

**Recommendation:** Collect and annotate a stratified Argentine Twitter/X sample (covering pampero, sudestada, AMBA flash flood, ENSO drought, and villa-impact events) using the CrisisBench label schema; use this for fine-tuning and as a regional held-out evaluation set. Apply TREC-IS-style protocols [WEB-15] and the cross-lingual transfer approach from arXiv:2209.02139 [WEB-17].

| ID     | Evidence                                                                                                                                                                                                                                        |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WEB-15 | TREC-IS introduced priority/criticality labels (Low/Medium/High/Critical) as the closest existing benchmark precedent for credibility scoring; covers no Argentine events ( <a href="https://trecis.github.io/">https://trecis.github.io/</a> ) |
| WEB-17 | Sánchez et al. 2022 cross-lingual crisis classification paper does not address Rioplatense register ( <a href="https://arxiv.org/abs/2209.02139">https://arxiv.org/abs/2209.02139</a> )                                                         |